

The Complete Guide to Shavian, Tengwar, and Featural Transliteration

Part 1: The Shavian Alphabet & ReadLex Specification

The Shavian alphabet is a purely phonemic writing system designed by Kingsley Read to fulfill the bequest of George Bernard Shaw. It replaces traditional non-phonetic English spelling with a 1:1 sound-to-letter mapping.

1. Character Taxonomy

Shavian characters fall into four structural categories based on vertical height and composition:

- **Tall (Voiceless Consonants):** Extend above the x-height.
- **Deep (Voiced Consonants):** Extend below the baseline.
- **Short (Vowels, Liquids, Nasals):** Fit entirely within the x-height.
- **Compound (Vocalic Ligatures):** Combine rhotic (/r/) sounds and vowel clusters into single glyphs.

Consonant Sound Pairs

Tall and Deep consonants form physical pairs where rotating or flipping a Tall (voiceless) letter generates its Deep (voiced) counterpart.

Sound Pair	Tall Glyph (Voiceless)	Deep Glyph (Voiced)	Phonetic Names
/p/ – /b/	ᵛ	ᵇ	peep – bib
/t/ – /d/	ᵀ	ᵀ̣	tot – dead
/k/ – /g/	ᵀ̣	ᵇ	kick – gag
/f/ – /v/	J	ᵀ̣	fee – vow
/θ/ – /ð/	ᵀ̣	ᵀ̣̣	thigh – they
/s/ – /z/	S	ᵀ̣̣	so – zoo

Sound Pair	Tall Glyph (Voiceless)	Deep Glyph (Voiced)	Phonetic Names
/ʃ/ – /ʒ/	ℓ	ɹ	sure – measure
/tʃ/ – /dʒ/	ℓ	ɹ	church – judge
/h/ – /w/	ɸ	/	he – woe

Vowels, Liquids, and Compound Ligatures

Category	Characters & Keywords
Liquids & Nasals	ℓ (/l/ loll), ʃ (/m/ mime), ʌ (/n/ nun), ɹ (/ŋ/ hung), ɔ (/r/ roar), ʌ (/j/ yea)
Short Vowels	ɪ (/ɪ/ if), ʌ (/ɛ/ egg), ʌ (/æ/ ash), ɹ (/ə/ ado), ɹ (/ʌ/ up), ʌ (/ɒ/ on), ʌ (/ʊ/ wool)
Long Vowels	ɪ (/i:/ eat), ɹ (/eɪ/ age), ɹ (/ɑ:/ ah), ɹ (/ɔ:/ awe), ʌ (/oʊ/ oak), ʌ (/u:/ ooze)
Diphthongs	ɹ (/aɪ/ ice), ɹ (/ɔɪ/ oil), ɹ (/aʊ/ out)
Compound Ligatures	ɹ (/ɑ:r/ are), ɹ (/ɔ:r/ or), ɹ (/ɛər/ air), ɹ (/ɜ:r/ err), ɹ (/ər/ array), ɹ (/ɪər/ ear), ɹ (/iə/ Ian), ʌ (/ju:/ yew)

2. The 12 ReadLex Orthographic Rules

- Strict Sound-to-Letter Phonetics:** Write strictly by spoken sound, ignoring silent Latin letters (e.g., *write* = ɹɹɹ).
- Phonetic Voicing Assimilation in Suffixes:** Suffixes match spoken voicing (e.g., *cats* = ʌɹɹ vs. *dogs* = ɹɹɹ; *walked* = ɹɹɹ vs. *played* = ɹɹɹ).
- The Big Five Abbreviations:** Five high-frequency function words are abbreviated as single shorthand letters: *the* = ʌ, *of* = ʃ, *and* = ʌ, *to* = 1, *for* = J.
- Monosyllabic Invariance:** Single-syllable words maintain their fully stressed spelling regardless of context (e.g., *but* is always ɹɹ).
- Monosyllabic Weak Vowel Exception:** Indefinite articles are explicitly written weak using schwa (ɹ): *a* = ɹ, *an* = ɹ.

6. **Stress Distinction in Polysyllabic Words:** Vowel reduction reflects stress shifts (e.g., noun *convict* = ɔ̃ʌʋɪɔ̃ vs. verb *convict* = ɔ̃ʌʋɪɔ̃).
 7. **No Letter Gemination:** Consonants are never doubled unless articulated twice across syllable boundaries (e.g., *annoy* = ʌʌ vs. *unnamed* = ʌʌʌʌ).
 8. **Syllabic Consonant Formatting:** Insert a schwa (ɪ) before terminal liquids or nasals (e.g., *little* = ɔ̃ʌʌɪ, *driven* = ɔ̃ʌʌʌ).
 9. **Verbal Contraction Formatting:** Negative contractions insert ɪ before ʌ and omit apostrophes (e.g., *didn't* = ɔ̃ʌʌʌ).
 10. **Proper Noun Marking:** Capital letters are omitted entirely; proper nouns use a middle Namer Dot (·) immediately preceding the word (e.g., *London* = ·ɔ̃ʌʌʌ).
 11. **Unstressed Terminal Vowels (happy Rule):** Words ending in unstressed -y or -ie use ɪ (e.g., *happy* = ʌʌʌ). Tense, fully stressed final vowels use ɪ (e.g., *trustee* = ʌʌʌʌ).
 12. **Cross-Dialectal "Rhotic RP" Standard:** Standard spelling uses a broad rhotic Received Pronunciation base so rhotic and non-rhotic speakers share a single format.
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Part 2: Tolkien's Tengwar & Everson 2001 (Alcarin) Standard

Tengwar is a featural abugida created by J.R.R. Tolkien. Primary consonants (*tengwar*) represent structural articulation features, while vowels (*tehtar*) sit above or below consonant bases as diacritics.

1. The Fëanorian Consonant Grid (Appendix E Layout)

In Tolkien's standard layout, primary consonants are organized into **4 Series** (Columns: Place of Articulation) and **6 Grades** (Rows: Manner of Articulation and Voicing):

- **Series I (Témar 1):** Dental / Alveolar (/t/, /d/, /θ/, /ð/)
- **Series II (Témar 2):** Labial / Bilabial (/p/, /b/, /f/, /v/)
- **Series III (Témar 3):** Palatal / Affricate (/tʃ/, /dʒ/, /ʃ/, /ʒ/)
- **Series IV (Témar 4):** Velar (/k/, /g/, /ŋ/)

Grade / Manner	Series I (Dental)	Series II (Labial)	Series III (Palatal)	Series IV (Velar)
Grade 1: Voiceless Stops (<i>Stem Down, Single Loop</i>)	ᵀ U+E000 <i>Tinco</i> (/t/)	ᵇ U+E001 <i>Parma</i> (/p/)	ɕ U+E002 <i>Calma</i> (/tʃ/)	ɟ U+E003 <i>Quesse</i> (/k/)
Grade 2: Voiced Stops (<i>Stem Down, Double Loop</i>)	ᵀ̥ U+E004 <i>Ando</i> (/d/)	ᵇ̥ U+E005 <i>Umbar</i> (/b/)	ɕ̥ U+E006 <i>Anga</i> (/dʒ/)	ɟ̥ U+E007 <i>Ungwe</i> (/g/)
Grade 3: Voiceless Fricatives (<i>Stem Up, Single Loop</i>)	ᶲ U+E008 <i>Thúle</i> (/θ/)	ᶂ U+E009 <i>Formen</i> (/f/)	ɕ̥ U+E00A <i>Harma</i> (/ʃ/)	ɟ̥ U+E00B <i>Hwesta</i> (/hw/)
Grade 4: Voiced Fricatives (<i>Stem Up, Double Loop</i>)	ᶲ̥ U+E00C <i>Anta</i> (/ð/)	ᶂ̥ U+E00D <i>Ampa</i> (/v/)	ɕ̥̥ U+E00E <i>Anca</i> (/ʒ/)	ɟ̥̥ U+E00F <i>Unque</i> (/gh/)
Grade 5: Nasals (<i>Short Stem, Double Loop</i>)	ᵐ U+E010 <i>Númen</i> (/n/)	ᵐ̥ U+E011 <i>Malta</i> (/m/)	ɕ̥̥̥ U+E012 <i>Noldo</i> (/ɲ/)	ɟ̥̥̥ U+E013 <i>Nwalme</i> (/ɳ/)
Grade 6: Glides / Semi-vowels (<i>Short Stem, Single Loop</i>)	ɹ̥ U+E014 <i>Óre</i> (/r/ final)	ɰ̥ U+E015 <i>Vala</i> (/w/)	ɕ̥̥̥̥ U+E016 <i>Anna</i> (/j/)	ɟ̥̥̥̥ U+E017 <i>Vilya</i>

2. Additional & Irregular Tengwar (Everson 2001 / Alcarin Standard)

Beyond the standard 24-tengwa grid, specialized base characters handle liquids, sibilants, glottal sounds, and silent carriers:

Tengwa Glyph	Code Point	Tengwa Name	Structural Description / Function
ȝ	U+E020	<i>Rómen</i>	Trilled/prevocalic /r/ (used before vowels)
Ƨ	U+E022	<i>Lamba</i>	Upward hooked liquid character (/l/)
ᄁ	U+E024	<i>Silme</i>	Upright S-curve sibilant (/s/)
ᄂ	U+E025	<i>Silme Nuquerna</i>	Inverted S-curve (used when carrying top tehta)
ᄃ	U+E026	<i>Esse / Aze</i>	Z-curve / inverted double loop (/z/)
ᄄ	U+E027	<i>Esse Nuquerna</i>	Inverted Z-curve (used when carrying top tehta)
λ	U+E028	<i>Hyarmen</i>	Downward stem with left hook (/h/)
ǀ	U+E02C	Ára (Long Carrier)	Descending vertical stem (dotless 'j', drops below baseline) carrying long orphan vowels
ǁ	U+E02D	<i>Extended Carrier</i>	Ascending vertical stem (starts above x-height)
ǂ	U+E02E	Telco (Short Carrier)	Neutral x-height vertical stem (dotless 'i') carrying short orphan vowels

3. Vowel Tehtar & Diacritic Marks

In English General Mode, vowel diacritics (*tehtar*) sit directly above or below the following consonant base glyph:

Tehta Mark	Code Point	Tehta Name	Phonetic Mapping	Visual Placement
˙	U+E040	<i>a-tehta</i>	/æ/, /ɑː/, /aɪ/	Three dots in a triangle above base

Tehta Mark	Code Point	Tehta Name	Phonetic Mapping	Visual Placement
·	U+E044	i-tehta	/ɪ/, /iː/	Single dot above base
.	U+E045	schwa-tehta	/ə/ (schwa)	Single dot below base
´	U+E046	e-tehta	/ɛ/, /eɪ/	Single acute stroke above base
,	U+E047	long e-tehta	/iː/ (FLEECE)	Double acute stroke above base
ˆ	U+E04A	o-tehta	/ɒ/, /ɔː/, /oʊ/	Right-facing curl above base
˘	U+E04C	u-tehta	/ʌ/, /ʊ/, /uː/	Left-facing curl above base
˙	U+E04D	long u-tehta	/uː/ (GOOSE)	Double left-facing curl above base
ˉ	U+E050	nasal bar	Preconsonantal nasal	Horizontal tilde above base stop

Part 3: The Shavian-to-Tengwar Transliteration Bridge

By using Shavian as an Intermediate Representation (IR), transliteration bypasses non-phonetic English orthography entirely, converting text through a 5-pass state machine.

1. Direct Phoneme Mapping Matrix

IPA Phoneme	Shavian Glyph	Tengwa	Target Tengwa / Tehta	Code Point Stream	State Machine Transformation Rule
/p/	ṗ	ṑ	Parma	U+E001	Direct 1:1 Consonant Base
/b/	ḃ	Ḃ	Umbar	U+E005	Direct 1:1 Consonant Base
/t/	ṭ	ṑ	Tinco	U+E000	Direct 1:1 Consonant Base

IPA Phoneme	Shavian Glyph	Tengwa	Target Tengwa / Tehta	Code Point Stream	State Machine Transformation Rule
/d/	ᄀ	ᄁ	<i>Ando</i>	U+E004	Direct 1:1 Consonant Base
/k/	ᄃ	ᄄ	<i>Quesse</i>	U+E003	Direct 1:1 Consonant Base
/g/	ᄅ	ᄆ	<i>Ungwe</i>	U+E007	Direct 1:1 Consonant Base
/f/	ᄇ	ᄈ	<i>Formen</i>	U+E009	Direct 1:1 Consonant Base
/v/	ᄉ	ᄊ	<i>Ampa</i>	U+E00D	Direct 1:1 Consonant Base
/θ/	ᄌ	ᄍ	<i>Thúle</i>	U+E008	Direct 1:1 Consonant Base
/ð/	ᄎ	ᄏ	<i>Anta</i>	U+E00C	Direct 1:1 Consonant Base
/s/	ᄐ	ᄑ	<i>Silme / Nuquerna</i>	U+E024 / U+E025	Flips to <i>Silme Nuquerna</i> (U+E025) if top tehta present
/z/	ᄒ	ᄓ	<i>Esse / Nuquerna</i>	U+E026 / U+E027	Flips to <i>Esse Nuquerna</i> (U+E027) if top tehta present
/ʃ/	ᄔ	ᄕ	<i>Harma</i>	U+E00A	Direct 1:1 Consonant Base
/ʒ/	ᄖ	ᄌ	<i>Anca</i>	U+E00E	Direct 1:1 Consonant Base
/tʃ/	ᄘ	ᄄ	<i>Calma</i>	U+E002	Direct 1:1 Consonant Base
/dʒ/	ᄚ	ᄆ	<i>Anga</i>	U+E006	Direct 1:1 Consonant Base
/j/	ᄜ	ᄡ	<i>Anna</i>	U+E016	Direct 1:1 Consonant Base
/w/	ᄞ	ᄢ	<i>Vala</i>	U+E015	Direct 1:1 Consonant Base
/hw/	ᄠ	ᄤ	<i>Hwesta</i>	U+E00B	Lookahead intercept:

IPA Phoneme	Shavian Glyph	Tengwa	Target Tengwa / Tehta	Code Point Stream	State Machine Transformation Rule
					Aspirated Wh cluster to <i>Hwesta</i>
/ŋ/	𐌵	𐌶	<i>Nwalme</i>	U+E013	Direct 1:1 Consonant Base
/h/	𐌷	𐌸	<i>Hyarmen</i>	U+E028	Direct 1:1 Consonant Base
/l/	𐌹	𐌺	<i>Lamba</i>	U+E022	Direct 1:1 Consonant Base
/m/	𐌻	𐌼	<i>Malta</i>	U+E011	Direct 1:1 Consonant Base
/n/	𐌽	𐌾	<i>Númen</i>	U+E010	Direct 1:1 Consonant Base
/r/	𐌿	𐌿 / 𐍀	<i>Rómen / Óre</i>	U+E020 / U+E014	Contextual: <i>Rómen</i> before vowels; <i>Óre</i> word-final/consonantal
/ɪ/	𐍁	𐍂	<i>i-tehta</i>	U+E044	Attach dot above next available consonant
/i:/	𐍃	𐍄	<i>Ára + i-tehta</i>	U+E02C U+E044	FLEECE vowel: Attach to Long Carrier or double acute on base
/ɛ/	𐍅	𐍆	<i>e-tehta</i>	U+E046	Attach acute stroke above next available consonant
/eɪ/	𐍇	𐍈	<i>Anna + e-tehta</i>	U+E016 U+E046	FACE diphthong: Offglide Anna base + <i>e-tehta</i>
/æ/	𐍉	𐍊	<i>a-tehta</i>	U+E040	Attach three dots above next available consonant
/aɪ/	𐍋	𐍌	<i>Anna + a-tehta</i>	U+E016 U+E040	PRICE diphthong: Offglide Anna base + <i>a-tehta</i>

IPA Phoneme	Shavian Glyph	Tengwa	Target Tengwa / Tehta	Code Point Stream	State Machine Transformation Rule
/ə/	ṙ	!	<i>schwa-tehta</i>	U+E045	Attach single dot below next available consonant
/ʌ/	ṛ	ı̇	<i>u-tehta</i>	U+E04C	STRUT vowel: Attach left curl above next available consonant
/ɒ/	Ṛ	ı̇	<i>o-tehta</i>	U+E04A	LOT vowel: Attach right curl above next available consonant
/oʊ/	o	ṛ̇	<i>Vala + o-tehta</i>	U+E015 U+E04A	GOAT diphthong: Offglide <i>Vala</i> base + <i>o-tehta</i>
/ʊ/	v	ı̇	<i>foot-tehta</i>	U+E04C	FOOT vowel: Attach left curl above next available consonant
/uː/	ʌ	ı̇	<i>Ára + u-tehta</i>	U+E02C U+E04C	GOOSE vowel: Attach to Long Carrier or double curl on base
/aʊ/	◀	ṛ̇	<i>Vala + a-tehta</i>	U+E015 U+E040	MOUTH diphthong: Offglide <i>Vala</i> base + <i>a-tehta</i>
/ɔɪ/	▷	á	<i>Anna + o-tehta</i>	U+E016 U+E04A	CHOICE diphthong: Offglide <i>Anna</i> base + <i>o-tehta</i>
/ɑː/	s	ı̇	<i>a-tehta</i>	U+E040	PALM vowel: Attach three dots above next available consonant

IPA Phoneme	Shavian Glyph	Tengwa	Target Tengwa / Tehta	Code Point Stream	State Machine Transformation Rule
/ɔɪ/	᳚	í	<i>awe-tehta</i>	U+E04A	THOUGHT vowel: Attach right curl above next available consonant
/iə/	᳛	á	<i>Anna + i-tehta</i>	U+E016 U+E044	IAN ligature: Offglide Anna base + <i>i-tehta</i>
/juɪ/	᳜	á	<i>Anna + u-tehta</i>	U+E016 U+E04C	YEW ligature: Offglide Anna base + <i>u-tehta</i>
/ɑɪr/	᳝	ȳ / ń	<i>Rómen / Óre + a-tehta</i>	U+E020/U+E014 U+E040	START vowel: Decomposes into R-base + <i>a-tehta</i>
/ɔɪr/	᳞	ȳ / ń	<i>Rómen / Óre + o-tehta</i>	U+E020/U+E014 U+E04A	NORTH vowel: Decomposes into R-base + <i>o-tehta</i>
/ər/	᳟	ȳ / ɳ	<i>Rómen / Óre + schwa-tehta</i>	U+E020/U+E014 U+E045	letter vowel: Decomposes into R-base + <i>schwa-tehta</i>
/ɪər/	᳠	ȳ / ń	<i>Rómen / Óre + i-tehta</i>	U+E020/U+E014 U+E044	NEAR vowel: Decomposes into R-base + <i>i-tehta</i>
/ɛər/	᳡	ȳ / ń	<i>Rómen / Óre + e-tehta</i>	U+E020/U+E014 U+E046	SQUARE vowel: Decomposes into R-base + <i>e-tehta</i>
/ɜɪr/	᳢	ȳ / ń	<i>Rómen / Óre + u-tehta</i>	U+E020/U+E014 U+E04C	NURSE vowel: Decomposes into R-base + <i>u-tehta</i>

2. The Big Five: Shorthand Logograms

Target Word	Shavian IR	Tengwar Output	Glyph Name	Code Point Stream	State Machine Rule
<i>the</i>	ø	þ	<i>Extended Anta</i>	U+E01C	Standalone logogram for "the"
<i>of</i>	ƿ	þ	<i>Extended Ampa</i>	U+E01D	Standalone logogram for "of"
<i>and</i>	ʌ	þ̄	<i>Ando + Nasal Bar</i>	U+E004 U+E050	Standalone shorthand for "and"
<i>to</i>	1	p	<i>Tinco Base</i>	U+E000	Unadorned base shorthand for "to"
<i>for</i>	J	h	<i>Formen Base</i>	U+E009	Unadorned base shorthand for "for"